OIPPP – Branch Coverage Report

Project	Online Internship & Placement Preparation Portal (OIPPP)
Issue	#98
Prepared by	E S Sukruth
Measured on	05-11-2026 (final rerun on the last commit on 06-11-2026, section 7)
Baseline commit	`8cb16a0` (`dev`, synced with `main`; release `v1.0.0` plus Part-3 test fixes)
After commit	`8cb16a0` plus the three test files added on branch `test/issue-98-branch-coverage` (tests only, no product code)
Environment	Local, Windows 11, Node.js 24.11.0, npm 11.6.2; server Jest 29.7.0 with mongodb-memory-server; client Vitest 5.0.5 with `@vitest/coverage-v8` (#90)

1. Summary

Component	Branches before	Branches after	Statements after	Tests
Server (`server/src`)	**89.22%** (555 / 622)	**94.37%** (587 / 622)	96.16% (1,254 / 1,304)	509 of 509 pass (43 suites)
Client (`client/src`)	33.06% (326 / 986)	33.06% (326 / 986)	38.36% (483 / 1,259)	53 of 53 pass (6 files)

- The issue quoted 89.06% (554 / 622) at `v1.0.0`. On `8cb16a0` it was 89.22% (555 / 622) before this work: `listing.search.service.js` had one more branch covered (26 / 30) by tests merged after the tag.
- 22 new server tests cover **32 of the 42 branches** that were missed in the eight files under 85%. All eight files are now at or above 84.6%, and seven are at or above 90%.
- The 10 branches still missed in those files are defensive or unreachable through the API (section 4). Covering them would need tests that call internal functions with impossible inputs, which proves nothing about the product.
- QA-4 asks for 70% server **statement** coverage. Branch coverage has no SRS or Test Plan target, so this report aims to test every real path and explain the rest.

How it was measured

```
# server
cd server
npx jest --runInBand --coverage --coverageReporters=json-summary --coverageReporters=json --
coverageReporters=text
# client (tooling from #90)
cd client
npx vitest run --coverage --coverage.reporter=json-summary --coverage.reporter=json --
coverage.reporter=text
```

Branches are counted by Istanbul (server) and V8 (client): each side of an `if`, `? :`, `&&`, `||`, `??` and each default parameter is one branch.

2. Server branch coverage by module

Module (`server/src`)	Before	After	Change
`config`	19 / 22 (86.4%)	19 / 22 (86.4%)	–
`middleware`	46 / 46 (100%)	46 / 46 (100%)	–
`modules/analytics`	59 / 71 (83.1%)	70 / 71 (98.6%)	+11
`modules/applications`	54 / 63 (85.7%)	59 / 63 (93.7%)	+5
`modules/audit`	4 / 4 (100%)	4 / 4 (100%)	–
`modules/listings`	79 / 92 (85.9%)	87 / 92 (94.6%)	+8
`modules/notifications`	20 / 27 (74.1%)	27 / 27 (100%)	+7
`modules/settings`	4 / 4 (100%)	4 / 4 (100%)	–
`modules/tests` (aptitude)	166 / 181 (91.7%)	166 / 181 (91.7%)	–
`modules/users`	103 / 110 (93.6%)	103 / 110 (93.6%)	–
`utils`	1 / 2 (50%)	2 / 2 (100%)	+1
Total	**555 / 622 (89.22%)**	**587 / 622 (94.37%)**	**+32**

3. Server branch coverage by file

Files with no branches (route files, `app.js`, `index.js`, `AppError.js`, `wrap.js`, `escape-regex.js`, `audit.service.js`, `auth.validation.js`, `application.deps.js`, `request-log.js`) are left out; they are at 100% statements in the Code Coverage Report.

File (`server/src`)	Before	After
`config/env.js`	19 / 22 (86.4%)	19 / 22 (86.4%)
`middleware/auth.js`	14 / 14 (100%)	14 / 14 (100%)
`middleware/error.js`	20 / 20 (100%)	20 / 20 (100%)
`middleware/rate-limit.js`	1 / 1 (100%)	1 / 1 (100%)
`middleware/validate.js`	11 / 11 (100%)	11 / 11 (100%)
`modules/analytics/analytics.service.js`	**59 / 71 (83.1%)**	70 / 71 (98.6%)
`modules/applications/application.constants.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/applications/application.model.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/applications/application.service.js`	**50 / 59 (84.7%)**	55 / 59 (93.2%)
`modules/audit/auditLog.model.js`	4 / 4 (100%)	4 / 4 (100%)
`modules/listings/bookmark.model.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/listings/bookmark.service.js`	**10 / 13 (76.9%)**	13 / 13 (100%)
`modules/listings/listing.model.js`	5 / 5 (100%)	5 / 5 (100%)
`modules/listings/listing.routes.js`	7 / 7 (100%)	7 / 7 (100%)
`modules/listings/listing.search.service.js`	**26 / 30 (86.7%)** [1]	27 / 30 (90.0%)
`modules/listings/listing.service.js`	22 / 22 (100%)	22 / 22 (100%)
`modules/listings/recommendation.service.js`	**7 / 13 (53.8%)**	11 / 13 (84.6%)
`modules/notifications/mailer.js`	**3 / 6 (50%)**	6 / 6 (100%)
`modules/notifications/notification.model.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/notifications/notification.service.js`	**15 / 19 (78.9%)**	19 / 19 (100%)

File (`server/src`)	Before	After
`modules/settings/settings.model.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/settings/settings.service.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/tests/aptitude.routes.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/tests/aptitude.service.js`	84 / 95 (88.4%)	84 / 95 (88.4%)
`modules/tests/attempt-timeout.job.js`	6 / 6 (100%)	6 / 6 (100%)
`modules/tests/attempt.model.js`	37 / 40 (92.5%)	37 / 40 (92.5%)
`modules/tests/question.model.js`	10 / 10 (100%)	10 / 10 (100%)
`modules/tests/result.model.js`	20 / 21 (95.2%)	20 / 21 (95.2%)
`modules/tests/schema.helpers.js`	5 / 5 (100%)	5 / 5 (100%)
`modules/tests/test.model.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/users/admin.service.js`	24 / 28 (85.7%)	24 / 28 (85.7%)
`modules/users/auth.service.js`	36 / 37 (97.3%)	36 / 37 (97.3%)
`modules/users/profile.model.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/users/profile.routes.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/users/profile.service.js`	14 / 14 (100%)	14 / 14 (100%)
`modules/users/public-user.js`	2 / 2 (100%)	2 / 2 (100%)
`modules/users/resume.storage.js`	10 / 11 (90.9%)	10 / 11 (90.9%)
`modules/users/resume.upload.js`	11 / 12 (91.7%)	11 / 12 (91.7%)
`modules/users/user.model.js`	2 / 2 (100%)	2 / 2 (100%)
`utils/logger.js`	**1 / 2 (50%)**	2 / 2 (100%)

Bold: the eight files under 85% listed in #98. [1] 25 / 30 (83.3%) at `v1.0.0`.

4. Missed branches in the eight files

Real means a user, an admin, the deployment or a concurrent request can take that path, so it gets a test. **Defensive** means the code guards against something that the routes, the validation layer or the schema already rule out; it stays, but a test would only call internals with impossible input.

4.1 `modules/notifications/mailer.js` (3 / 6 → 6 / 6)

Line	Branch	Kind	Test added
9–10	`smtp.host` set → real SMTP transport	Real: every deployed build (Mailtrap on the test build, Brevo in `render.yaml`)	`unit/mailer-logger-config.test.js` "with SMTP_HOST and a user it builds an authenticated SMTP transport once"
14	`smtp.user` set → `auth` block	Real: authenticated SMTP	same test
14	no `smtp.user` → `auth: undefined`	Real: an open relay or local SMTP; also checks port 465 → `secure: true`	"port 465 uses TLS; no SMTP_USER means no auth block"

4.2 `utils/logger.js` (1 / 2 → 2 / 2)

Line	Branch	Kind	Test added
4	`NODE_ENV !== 'test'` → level `info`	Real: production and development	`unit/mailer-logger-config.test.js` "is silent under test and info otherwise"

4.3 `modules/listings/recommendation.service.js` (7 / 13 → 11 / 13)

Line	Branch	Kind	Test added
8	no Profile → 404 `NOT_FOUND`	Real: a student who has not saved a profile yet	`integration/branch-coverage-listings.test.js` "a student without a profile gets 404"
34	profile without CGPA → only listings with no minimum	Real: CGPA is optional in the profile	"a profile without CGPA only sees listings with no CGPA minimum"
60–61	equal match count and equal deadline → order by `_id`	Real: two listings with the same skills and deadline (common in `seed:demo`)	"equal match count and deadline fall back to listing id order" (covers both arms)
13	`profile.skills` missing → `[]`	Defensive: the Profile schema defaults `skills` to `[]`	–
45	`listing.skills` missing → `[]`	Defensive: the Listing schema defaults `skills` to `[]`	–

4.4 `modules/listings/bookmark.service.js` (10 / 13 → 13 / 13)

Line	Branch	Kind	Test added
11	listing not found → 404	Real: a stale link or a deleted listing	`integration/branch-coverage-listings.test.js` "bookmarking a listing that does not exist is 404"
29	`create` hits the unique index (11000) → return the existing bookmark	Real: two quick clicks or two tabs (SRCH-4, no duplicates)	"a concurrent duplicate save (unique index 11000) returns the existing bookmark"
29	any other database error → rethrow	Real: a database failure must give 500, not a fake success	"any other database error on save is not swallowed"

4.5 `modules/notifications/notification.service.js` (15 / 19 → 19 / 19)

Line	Branch	Kind	Test added
14	recipient has no email → retry later	Real: the user record was removed or has no email	`integration/branch-coverage-analytics-applications.test.js` "a recipient with no email is marked for retry, not sent"
38	`retryDueEmails()` default `now`	Real: the 60 s job calls it with no argument	"defaults: retry uses the current time, list works without options, the job uses 60 s"
44	`startEmailRetryJob()` default 60,000 ms	Real: `index.js` starts the job with no argument	same test
52	`list(userId)` default options	Real: internal callers without a filter	same test

4.6 `modules/analytics/analytics.service.js` (59 / 71 → 70 / 71)

Line	Branch	Kind	Test added
26	equal percentage and equal size → fixed topic order (decision D2)	Real	`integration/branch-coverage-analytics-applications.test.js` "weak-topic ties on percentage and size fall back to the fixed topic order"
45	test record missing → `testTitle: null`	Real: the trend must not crash if a test is removed in the database	"a result whose test record is gone shows testTitle null, not a crash"
61, 63	profile without `branch` → `branch: null`, unfiltered and year-only filter (decision D4)	Real: `branch` is optional in the profile	"profiles without a branch are reported as branch null, also under a year-only filter"
84, 86	listing missing → `company` / `title` null, kept in totals	Real: documented in analytics-api.md ("a listing that no longer exists is reported with `title` and `company` set to `null`, not dropped")	"offers on a deleted listing are kept as 'Unknown listing', in JSON and CSV, with branch rows"
145, 147	CSV `?? 'Unknown listing'` / `?? ''` for those rows	Real: same case in the export (ANL-3)	same test
150	CSV branch rows, named and `No branch`	Real: the existing CSV tests had no results, so no branch rows were ever written	same test
132	`csvCell(null)` → empty cell	Defensive: every caller passes a value or a `??` fallback; kept so a future column cannot print `null`	–

4.7 `modules/listings/listing.search.service.js` (26 / 30 → 27 / 30)

Line	Branch	Kind	Test added
63	`deadlineAfter` in the past → clamp to now	Real: a student picks an old "deadline after" date; expired listings must stay hidden (SRCH-2, LST-4)	`integration/branch-coverage-listings.test.js` "a deadlineAfter in the past still hides expired listings"
23, 24	default `page = 1`, `limit = 20`	Defensive: the Joi schema on `GET /listings` always fills both	–
66	unknown `sortBy` → `deadline`	Defensive: Joi rejects any other `sortBy` with 400 before the service runs	–

4.8 `modules/applications/application.service.js` (50 / 59 → 55 / 59)

Line	Branch	Kind	Test added
15	listing not found → 404	Real: `GET /listings/:id/applications` with a wrong ID	`integration/branch-coverage-analytics-applications.test.js` "applicants of a listing that does not exist is 404"
83	listing missing → `listing: null` in *My applications*	Real: the student keeps the application record (SAFE-2 spirit)	"a student's application whose listing was deleted still lists, with listing null"
109	applicant without a Profile → empty profile fields	Real: a student can apply before saving a profile when the listing has no CGPA minimum	"an applicant without a profile is listed with empty profile fields"

Line	Branch	Kind	Test added
174	application not found → 404 on its resume	Real: a stale link	"resume of an unknown application is 404"
182	non-recruiter (student owner, admin) skips the owner check	Real: SAD row 16, the owner and the admin may read the resume	"the student owner and an admin pass the access check for an application resume"
27	`alreadyApplied(null)`	Defensive: reached only if the duplicate-key error fires and the winning record vanishes before the read	–
60	`create` error other than 11000 → rethrow	Defensive: a database outage; the error middleware returns 500	–
88 (×2)	default `page = 1`, `limit = 20`	Defensive: Joi fills both on `GET /listings/:id/applications`	–

4.9 Totals for the eight files

	Branches
Missed before	42
Real paths, now tested	32
Defensive or unreachable, left as is	10
Missed after	10

5. Other server files above 85% but not 100%

Not in scope for #98, listed so nothing is hidden. Most are guards on configuration or rare database failures.

File	Missed	Notes
`modules/tests/aptitude.service.js`	11	Not-found and stale-version guards on edits, the 503 `RESULT_UNAVAILABLE` path for a legacy attempt with no result, a non-duplicate insert error, a concurrent `ATTEMPT_CHANGED`, the startup check that refuses a standalone MongoDB, and two role ternaries. The main paths have 30+ integration tests
`modules/users/admin.service.js`	4	Reject-reason guard (Joi already checks it), pagination defaults, unknown student on the admin resume route
`config/env.js`	3	`NODE_ENV` default and the production-only `JWT_SECRET` check
`modules/tests/attempt.model.js`	3	`answers
`modules/tests/result.model.js`, `auth.service.js`, `resume.storage.js`, `resume.upload.js`	1 each	`topicBreakdown

6. Client branch coverage

Client tests (#90 tooling) are page-level Vitest + Testing Library tests written for the riskiest screens (login, register, applicants, attempt, progress, settings). No client tests were added in #98: code is frozen and the server is where the business rules and branches are. The low client number reflects pages that are only exercised by system and UI tests (UI-TST-01 to UI-ANL-02, UAT in #97).

6.1 By folder

Folder (`client/src`)	Branches
`api`	32 / 38 (84.2%)
`auth`	9 / 12 (75.0%)
`components`	28 / 116 (24.1%)
`lib`	68 / 97 (70.1%)
`pages` (public)	32 / 48 (66.7%)
`pages/admin`	35 / 235 (14.9%)
`pages/recruiter`	64 / 145 (44.1%)
`pages/student`	58 / 291 (19.9%)
`App.jsx`	0 / 4 (0%)
Total	**326 / 986 (33.06%)**

6.2 By file

File	Branches	File	Branches
`api/client.js`	32 / 38 (84.2%)	`pages/admin/Questions.jsx`	12 / 70 (17.1%)
`auth/AuthContext.jsx`	3 / 6 (50%)	`pages/admin/Settings.jsx`	13 / 14 (92.9%)
`auth/RequireRole.jsx`	6 / 6 (100%)	`pages/admin/Statistics.jsx`	0 / 43 (0%)
`components/Layout.jsx`	0 / 20 (0%)	`pages/admin/TestsAdmin.jsx`	10 / 62 (16.1%)
`components/ListingCard.jsx`	0 / 21 (0%)	`pages/admin/Users.jsx`	0 / 46 (0%)
`components/NotificationBell.jsx`	0 / 25 (0%)	`pages/recruiter/Applicants.jsx`	45 / 62 (72.6%)
`components/PdfViewer.jsx`	0 / 12 (0%)	`pages/recruiter/ListingForm.jsx`	19 / 57 (33.3%)
`components/ui.jsx`	28 / 38 (73.7%)	`pages/recruiter/MyListings.jsx`	0 / 26 (0%)
`lib/format.js`	10 / 17 (58.8%)	`pages/student/Applications.jsx`	13 / 28 (46.4%)
`lib/status.js`	3 / 4 (75%)	`pages/student/Attempt.jsx`	31 / 57 (54.4%)
`lib/timer.js`	16 / 18 (88.9%)	`pages/student/Home.jsx`	0 / 19 (0%)
`lib/useApi.js`	4 / 8 (50%)	`pages/student/ListingDetail.jsx`	0 / 46 (0%)
`lib/useBookmarks.js`	0 / 12 (0%)	`pages/student/Profile.jsx`	0 / 62 (0%)
`lib/validation.js`	35 / 38 (92.1%)	`pages/student/Progress.jsx`	14 / 18 (77.8%)
`pages/Forbidden.jsx`	1 / 2 (50%)	`pages/student/Result.jsx`	0 / 4 (0%)
`pages/Login.jsx`	11 / 21 (52.4%)	`pages/student/Saved.jsx`	0 / 4 (0%)
`pages/Register.jsx`	20 / 25 (80%)	`pages/student/Search.jsx`	0 / 35 (0%)
		`pages/student/Tests.jsx`	0 / 18 (0%)

`lib/chart.js`, `lib/download.js`, `main.jsx`, `pages/Help.jsx` and `pages/NotFound.jsx` have no branches.

The logic that decides correctness on the client (`lib/timer.js` 88.9%, `lib/validation.js` 92.1%, `api/client.js` 84.2%, `auth/RequireRole.jsx` 100%) is well covered. The 0% pages are display code; the next client tests worth writing are `Statistics.jsx`, `Tests.jsx` and `Result.jsx` (aptitude and analytics) and `Search.jsx` (QA-1 path).

7. Final rerun on the last commit (Fri 06-11-2026)

To be filled in before 12 pm on 06-11-2026, on the final `dev`/`main` commit after all Part-3 PRs are merged.

	Server branches	Server statements	Server tests	Client branches	Client tests
05-11-2026, `8cb16a0` + #98 tests	94.37% (587 / 622)	96.16% (1,254 / 1,304)	509 / 509	33.06% (326 / 986)	53 / 53
06-11-2026, final commit `_____`					

8. Conclusion

Server branch coverage rose from 89.22% to 94.37% with tests only. Every real path that was missed in the eight files under 85% now has a test, and each of the 10 branches left uncovered in them is named above with the reason it is defensive. All 509 server tests and 53 client tests pass. Client branch coverage is 33.06%; its gaps are display pages covered by the UI test cases and UAT rather than unit tests.